

PowerScan

Smart QR Electricity Bill Payment System — Project Specification

1. Project Idea

PowerScan is a digital electricity-bill payment system that connects an electricity meter to a customer's digital account using a unique QR code. The customer scans the QR code using the PowerScan application, the system identifies the associated meter/account, retrieves the bill, and provides a secure payment flow.

2. Problem

Customers may need to enter long consumer/service numbers when paying electricity bills. Manual entry can be inconvenient and can lead to typing mistakes. PowerScan aims to make the process simpler by using a QR code associated with the meter.

3. Proposed Solution

- Each electricity meter is associated with a unique QR code.
- The customer opens the PowerScan app and scans the QR code.
- The backend identifies the meter/account.
- The latest bill information is retrieved and displayed.
- The customer completes the payment.
- A digital payment receipt is generated.

4. Customer Flow

Open App → Scan QR → Identify Meter → Fetch Bill → Display Bill → Pay → Payment Confirmation → Digital Receipt

5. Main Features

- QR code scanner
- Automatic meter/account identification
- Current bill display
- Due-date display
- Units consumed and meter-reading information
- Payment interface
- Digital payment receipt
- Bill history
- Payment history
- Due-date reminders
- Electricity usage history and graphs
- Meter/service complaint option
- Power-outage reporting as a future module
- Admin dashboard

6. System Architecture

The basic architecture consists of a frontend application, a backend API, a relational database and a payment service.

Component	Purpose
Frontend	Customer interface, QR scanning, bill display and payment screens
Backend	Authentication, QR/account mapping, bill APIs, business logic and transaction processing
Database	Stores customers, meters, bills and payment records
Payment Service	Processes the payment through an approved payment provider
Admin Dashboard	Manages and monitors system data and transactions

7. Recommended Technology Stack

Layer	Technology
Frontend	HTML, CSS, JavaScript
Backend	Python + FastAPI
Database	MySQL
QR scanning	Browser/mobile QR scanning library
Payment	Sandbox/test payment gateway for prototype

Python + FastAPI is recommended for a semester prototype because it allows the backend and APIs to be developed relatively quickly. Java/Spring Boot is also technically possible.

8. Database Design

PowerScan uses a relational MySQL database. The main relationship pattern is one-to-many (1:N): Customer → Meter → Bills → Payments.

Table	Important fields
customers	customer_id (PK), name, mobile, email, address
meters	meter_id (PK), customer_id (FK), meter_number, qr_code
bills	bill_id (PK), meter_id (FK), previous_reading, current_reading, units_consumed, bill_amount, due_date, status
payments	payment_id (PK), bill_id (FK), amount, payment_date, transaction_id, payment_status

Primary keys uniquely identify records; foreign keys connect related records. A customer may have multiple meters, a meter may have many bills, and bills can have payment records.

9. Electricity Bill Calculation

Units consumed = Current meter reading – Previous meter reading. The applicable tariff/slab is then used to calculate energy charges. Fixed charges, taxes/duties, other applicable charges and adjustments may also be included. The actual tariff must come from the authorized electricity provider.

10. Payment and Receipt Flow

- Customer selects the bill to pay.
- System validates the bill and amount.
- Customer completes payment through the approved payment provider.
- Payment status is verified.
- Transaction details are stored.
- Digital receipt is generated.
- Bill status is updated.

11. QR Code Design

Each meter should have a unique identifier encoded in or referenced by the QR code. The QR code should not directly expose sensitive personal information. The backend should validate the QR identifier and return only the information the authenticated customer is allowed to see.

12. Security Requirements

- HTTPS/encrypted communication
- Secure authentication
- Authorization checks for customer data
- Input validation
- Secure API design
- Payment verification
- Transaction logging
- Protection against QR copying and misuse
- Rate limiting and abuse prevention
- Regular database backups

13. Website/App Screens

- Home / Login
- Customer Dashboard
- QR Scanner
- Bill Details
- Payment
- Payment Success
- Digital Receipt
- Bill History
- Usage History
- Service/Complaint Page
- Profile
- Admin Dashboard

14. Semester Prototype Scope

For a college prototype, use simulated customer, meter and bill data. Demonstrate the complete workflow without connecting to a real electricity-provider database or real government account.

- Create MySQL tables and sample records.
- Generate unique QR codes for demo meters.
- Scan a QR code from the website/app.
- Fetch the corresponding demo bill through FastAPI.
- Display bill amount, units and due date.
- Use a simulated or sandbox payment.

- Generate a digital receipt.
- Add bill history and a basic usage chart.
- Create an admin dashboard.
- Test the complete workflow.

15. Example Prototype Flow

QR Code → Meter ID: MTR1001 → Customer: Demo Customer → Current Bill: ■1,250 → Due Date: 15/10/2026 → Pay → Payment Successful → Receipt Generated

16. Future Scope

- Smart-meter integration
- Automatic bill retrieval
- Real-time usage information
- Notifications and reminders
- Outage reporting
- Meter-service requests
- Advanced usage analytics
- Multi-provider support

17. Project Goal

The goal of PowerScan is to create a simple, secure and convenient digital electricity-service platform in which the physical meter becomes the easy entry point for identifying the customer's account, viewing the bill and completing payment.

Tagline: **Scan. Pay. Done.**